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Prop convergencia imp cauchy

Proposición 1. *Sea X un espacio métrico y $(x_n)_{n\in\mathbb{N}} \in X^{\mathbb{N}}$, entonces*

$$\exists x \in X : x_n \xrightarrow{n\to\infty} x \implies (x_n)_{n\in\mathbb{N}} \text{ es una sucesión de Cauchy.}$$

***Demostración*:** Sea $\varepsilon > 0$, como $x_n \xrightarrow{n\to\infty} x$, $\exists n_0 \in \mathbb{N} : \forall n \geqslant n_0 : d(x_n, x) < \varepsilon/2$. Entonces, por la desigualdad triangular, tenemos que

$$\forall n, m \geqslant n_0 : d(x_n, x_m) \leqslant d(x_n, x) + d(x_m, x) < \varepsilon.$$

Por tanto, $(x_n)_{n\in\mathbb{N}}$ es una sucesión de Cauchy. ■

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